

Nitin Sharma

 Portfolio |  Nitin Sharma |  nitin18geo@gmail.com |  (+91) 920-5646-560

SUMMARY

Geospatial analyst with 3 years of experience in water, land use, and agricultural applications. Skilled in programming (GEE, python, R), geospatial modelling, and water accounting. Experienced with decision-support tools for government stakeholders. I am available to join from August, 2026.

WORK EXPERIENCE

Young Professional (Consultant, Full-time) — School of Public Policy, IIT Delhi Feb 2025 – Present

- Spatial DSS for river rejuvenation (NMCG): Built simulation model for urban water management; and fed into a spatial DSS adopted by NMCG (*Vensim, QGIS, python Django*).
- Mapping equity, efficiency, and sustainability in agricultural water use for horticulture: Led project cycle: methodology design, indicator development, field data collection, and geospatial analysis pipeline (*GEE, Python*).

Research Consultant (Full-time) — IWMI, Delhi, India Jan 2024 – Dec 2024

- Evaluation of Direct Seeded Rice scheme, Haryana (*Ministry of Jal Shakti*): built ML-based crop classification using Sentinel SAR/Optical imagery and field data (*GEE*); planned drone mapping mission for field-level ET estimation (*Cloud-Compare (stitching), Python, QGIS*).
- Digital Innovation for Water-Secure Africa (global project): developed automated flood mapping pipelines using SAR and optical data across African countries (*GEE, Python, and Sandbox environment*); conducted development, QA/QC, and validation of drought/flood indicators (*Django Python*); produced geospatial risk maps for international disaster response (*Python, QGIS*).

Research Intern (Full-time) — IWMI, Delhi, India Jun 2023 – Dec 2023

- Built Water Storage for South Asia, river basin scale (*USAID-funded*): developed GEE workflows for surface water extent & storage estimation; and conducted accuracy assessment on 10m LULC global datasets, developed pipelines for automation of data scraping, processing and analysis. (*GEE, R*);
- Wrote StoryMaps article featured on CGIAR (*ArcGIS storyMaps*);(*Python*); delivered Google Earth Engine training to government agencies (CWC-Delhi, WRD Telangana).

Research Intern (Remote) — Arizona State University, AZ Jun 2023 – Aug 2023

- Prepared geospatial training datasets; conducted watershed-scale environmental risk assessment (*ArcGIS pro, Python*).

EDUCATION

2021 – 2023	Indian Institute of Technology , Kharagpur, India Master of Science — Geology Relevant courses: Remote Sensing & GIS	(GPA: 8.95, First Class)
2022 – 2023	National Law University and World Wide Forum – India , Remote P.G. Diploma — Environmental Law and Policy	(Grade: A)
2018 – 2021	Hansraj College, University of Delhi , Delhi, India Bachelor of Science (Hons.) — Geology Relevant courses: Remote Sensing & GIS	(GPA: 8.65, First Class)

SKILLS

Geospatial & Remote Sensing	GEE (scripting & app), QGIS (with PyQGIS and cloud-native), SAR/Optical datasets, time-series analysis, drone/UAV data collection
Programming & Modelling	Python (GeoPandas, Rasterio, (rio)Xarray), R, ML-classification pipeline, Geospatial modelling
Tools Development	Decision-support dashboards, GEE app development, database & data management (post-GIS), QA/QC
Field & Coordination	Field data collection, Stakeholder engagement, inter-team communication
Communication	Technical/research report writing, StoryMaps

CERTIFICATIONS

- Workshop on UAV/Drone Applications in Environment and NRM: Hands-on — TERI, 2024
- Spatial Analytics — ESRI, 2022
- Python for Data Science — NPTEL, 2019

PUBLISHED WORK

Journal Article:

- *Patterns of surface water dynamics and storage changes in a basin of Bundelkhand Region, India* — Journal of Irrigation and Drainage (Wiley), Nov 2024. [Link](#)

Articles/Reports:

- Building resilient river basins: the catchment risk and resilience (CRI) index in India. [Link](#)
- Effect of Rainfall Deficit on Surface Water Storage Changes. [CGIAR Article](#) [StoryMap](#)

Conference:

- Surface Water Mapping in Un-monitored Small Waterbodies, Ramganga Basin — AGU Hydrology 2024. [Abstract](#)

HONORS & AWARDS

- Best StoryMaps Award, ESRI India StoryMaps Contest, 2022 [Link](#)
- L-Austin Weeks Undergraduate Grant, 2020

PROFESSIONAL REFERENCES

- **Dr. Amarnath Giriraj** — Research Group Leader, Water Data for Climate Resilience, *IWMI, Sri Lanka*: a.giriraj@cgiar.org
- **Mr. Dhyey Bhatpuria** — Environment Data Researcher, *IWMI, Delhi, India*: d.bhatpuria@cgiar.org
- **Dr. Pooja Prasad** — Assistant Professor, School of Public Policy, *IIT Delhi*: p-pooja@iitd.ac.in